

Module 06: Relationships, Cardinalities & Weak Entities

1. Architectural Architecture Overview



2. Mapping Cardinalities & Participation Constraints

Ratio	Constraint Meaning	Total vs Partial Participation	Example
1 : 1	Citizen has at most 1 Passport	Passport has Total Participation	Citizen ↔ Passport
1 : N	1 Dept employs N Employees	Employee has Total (Must have a Dept)	Department → Employee
M : N	M Students enroll in N Courses	Both sides can be Partial/Total	Student ↔ Course

3. Weak Entity Set Rules (Exam Must-Know)

1. **Double Rectangle:** Weak entity container.

2. **Double Diamond:** Identifying relationship with Strong owner entity.

3. **Dashed Underline:** Partial key / Discriminator.

4. **Double Line:** Always total participation.

5. **Composite PK:** PK(Weak) = PK(Owner) + Discriminator.